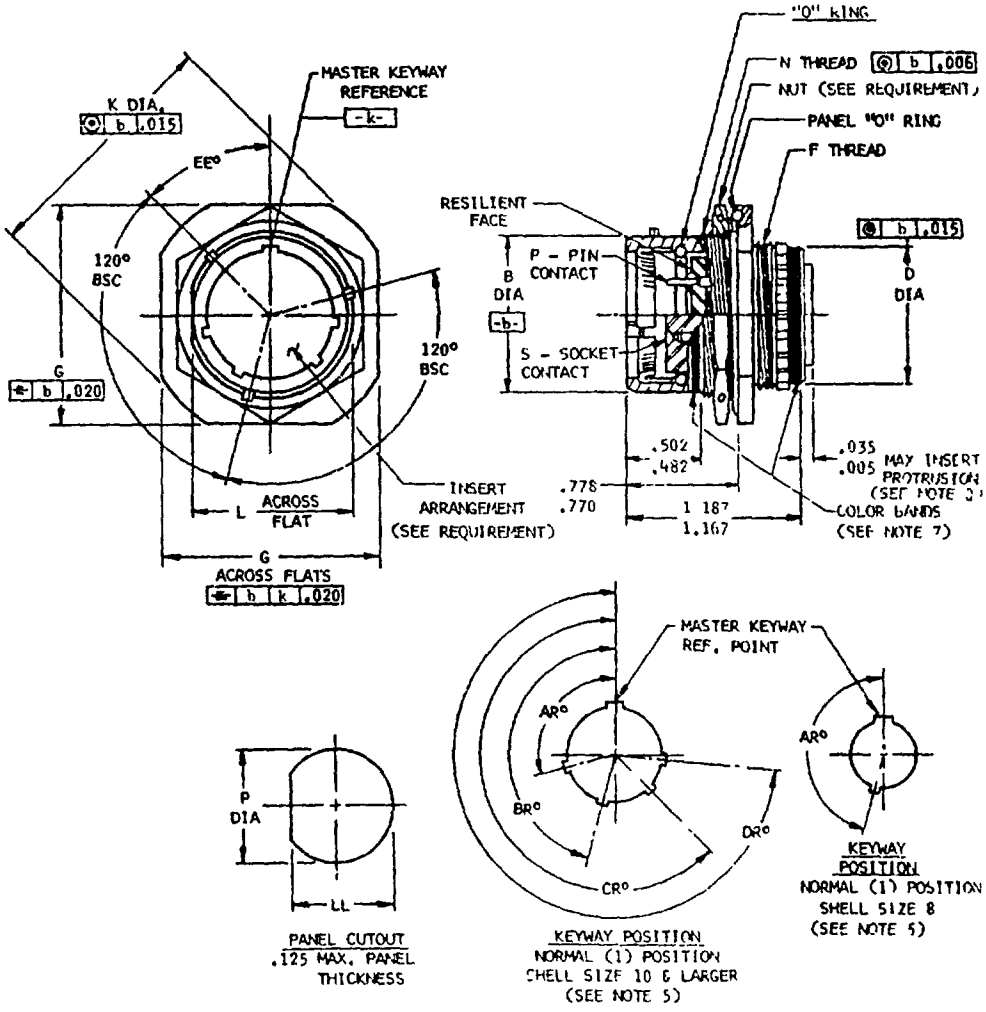
MILITARY SPECIFICATION SHEET

Ⓐ CONNECTORS, ELECTRICAL, CIRCULAR, HIGH DENSITY, QUICK DISCONNECT, ENVIRONMENT RESISTING: RECEPTACLE, JAM NUT MOUNT, INDIVIDUAL RELEASE, CRIMP-TYPE CONTACTS, CLASS A, F AND W (SERIES 4)

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the connectors described herein shall consist of this document and the latest issue of Specification MIL-C-81511.

THIS SPECIFICATION INACTIVE FOR NEW DESIGN AFTER 30 APRIL 1975. SEE MIL-STD-1353



Ⓐ denotes changes

FSC 5935

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TABLE I.

SHELL SIZE	B DIA	D DIA	F THREAD	G ACROSS FLATS	K DIA	L ACROSS FLATS	N THREAD	P DIA	LL
8	.553 (14.05) .547 (13.89)	.430 (10.92) .424 (10.77)	1/2-28 UNEF-2A	.989 (25.12) .969 (24.61)	1.072 (27.23) 1.052 (26.72)	.598 (15.19) .590 (14.99)	5/8-20 UN-2A	.640 (16.26) .630 (16.00)	.610 (15.49) .600 (15.24)
10	.678 (17.22) .672 (17.07)	.555 (14.10) .549 (13.94)	5/8-28 UN-2A	1.114 (28.30) 1.094 (27.79)	1.197 (30.40) 1.177 (29.90)	.723 (18.36) .715 (18.16)	3/4-20 UNEF-2A	.765 (19.43) .755 (19.18)	.735 (18.67) .725 (18.42)
14	.928 (23.57) .922 (23.42)	.805 (20.45) .799 (20.29)	7/8-28 UN-2A	1.401 (35.59) 1.381 (35.08)	1.510 (38.35) 1.490 (37.85)	.973 (24.71) .965 (24.51)	1-20 UNEF-2A	1.015 (25.78) 1.005 (25.53)	.985 (25.02) .975 (24.77)
16	1.054 (26.77) 1.048 (26.62)	.930 (23.62) .924 (23.47)	1-28 UN-2A	1.526 (38.76) 1.506 (38.25)	1.653 (41.53) 1.615 (41.02)	1.098 (27.89) 1.090 (27.69)	1 1/8-20 UN-2A	1.140 (28.96) 1.130 (28.70)	1.110 (28.19) 1.100 (27.94)
18	1.178 (29.92) 1.172 (29.77)	1.055 (26.80) 1.049 (26.64)	1 1/8-28 UN-2A	1.651 (41.94) 1.631 (41.43)	1.745 (44.32) 1.725 (43.82)	1.222 (31.04) 1.214 (30.84)	1 1/4-20 UN-2A	1.265 (32.13) 1.255 (31.88)	1.235 (31.37) 1.225 (31.12)

SHELL SIZE	AR° BSC	BR° BSC	CR° BSC	DR° BSC	PANEL "O" RING	EE° BSC - POSITION 1	2	3	4	5	6
8	128°	—	—	—	MS9068-018	40	40	40	40	40	40
10	105°	140°	215°	265°	MS9068-020	60	46	46	60	89	60
14	105°	140°	215°	265°	MS9068-024	60	46	46	60	89	60
16	105°	140°	215°	265°	MS9068-026	50	50	50	50	50	50
18	105°	140°	215°	265°	MS9068-028	50	50	50	50	50	50

INCHES	MM	INCHES	MM	INCHES	MM
.003	.08	.090	2.29	.502	12.75
.005	.13	.094	2.39	.770	19.56
.006	.15	.100	2.54	.778	19.76
.015	.38	.125	3.18	1.167	29.64
.020	.51	.200	5.08	.187	30.15
.067	1.70	.230	5.84	.035	.89
.073	1.85	.482	12.24		

NOTES:

1. DIMENSIONS ARE IN INCHES.
2. MILLIMETERS ARE IN PARENTHESES.
3. METRIC EQUIVALENTS (TO THE NEAREST .01 MM) ARE GIVEN FOR GENERAL INFORMATION ONLY AND ARE BASED UPON 1 INCH = 25.4 MM.
4. DIMENSIONS ARE AFTER PLATING.
5. NORMAL MATING KEYWAY POSITION SHOWN. FOR ALTERNATE KEYWAY POSITIONS SEE MIL-C-81511, FIGURE 4.
6. FOR SHELL SIZES 20, 22, AND 24 USE MIL-C-81511/23 OR MIL-C-81511/49 CONNECTORS.
7. COLOR BANDS: COLOR BLUE .100 MAX. WIDE, MUST BE VISIBLE WHEN MOUNTED.
8. ARRANGEMENTS CONTAINING SIZE 12 CONTACTS, INSERT PROTRUSION .070/.020.

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REQUIREMENTS:

Dimensions and configuration: See figure and table I.

Series: 4.

Ⓐ Classes: A, F and W.

Shell size: See table II.

Insert arrangement See supplement for MS sheet corresponding to desired shell size

Contacts: STyle P or S, crimp type

Shell polarization: 1 thru 6, key position.

Weight. See table II.

Marking: Part number on shell.

Mating counterpart· Plug connectors specified in MIL-C-81511/6, /34, and /56.

Insertion and removal tool. As specified in MS3160.

Slotted nut: Shall conform to MIL-C-81511/20.

Part number: The part number shall consist of M81511/53 together with the applicable codes for class, shell size, insert arrangement, contact style, and key positions. Information concerning the formulation of the part number can be found in MIL-C-81511.

TABLF II. Shell size and weight.

Shell size	Weight (ref) 1/ (pounds)
18 (Code F) 16 (Code E) 14 (Code D) 10 (Code B) 8 (Code A)	To be determined

1/ Removable crimp contacts not included in weight. (See contact drawings for weight of pin and socket contact.)

Custodian:
Navy - AS

Review activities:
Navy - EC, OS, SH
DSA - ES

User activity:
Navy - MC

Preparing activity
Navy - AS

Agent:
DSA - ES

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No. 22-R255

INSTRUCTIONS. The purpose of this form is to solicit beneficial comments which will help achieve procurement of suitable products at reasonable cost and minimum delay, or will otherwise enhance use of the document. DoD contractors, government activities, or manufacturers/vendors who are prospective suppliers of the product are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity.

DOCUMENT IDENTIFIER AND TITLE
MIL-C-81511/53A(Navy) Connectors, Electrical, Circular, High Density, Quick Disconnect

NAME OF ORGANIZATION AND ADDRESS

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY DOCUMENT REQUIREMENT CONSIDERED TOO RIGID

3. IS THE DOCUMENT RESTRICTIVE?

☐ YES ☐ NO *(If "Yes", in what way?)*

4 REMARKS

SUBMITTED BY *(Printed or typed name and address - Optional)*

TELEPHONE NO.

DATE

DD FORM 1 JAN 72 1426

REPLACES EDITION OF 1 JAN 66 WHICH MAY BE USED

S/N 0102-014-1802

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